

CS223 Computer Architecture & Organization

Neural Network Accelerators



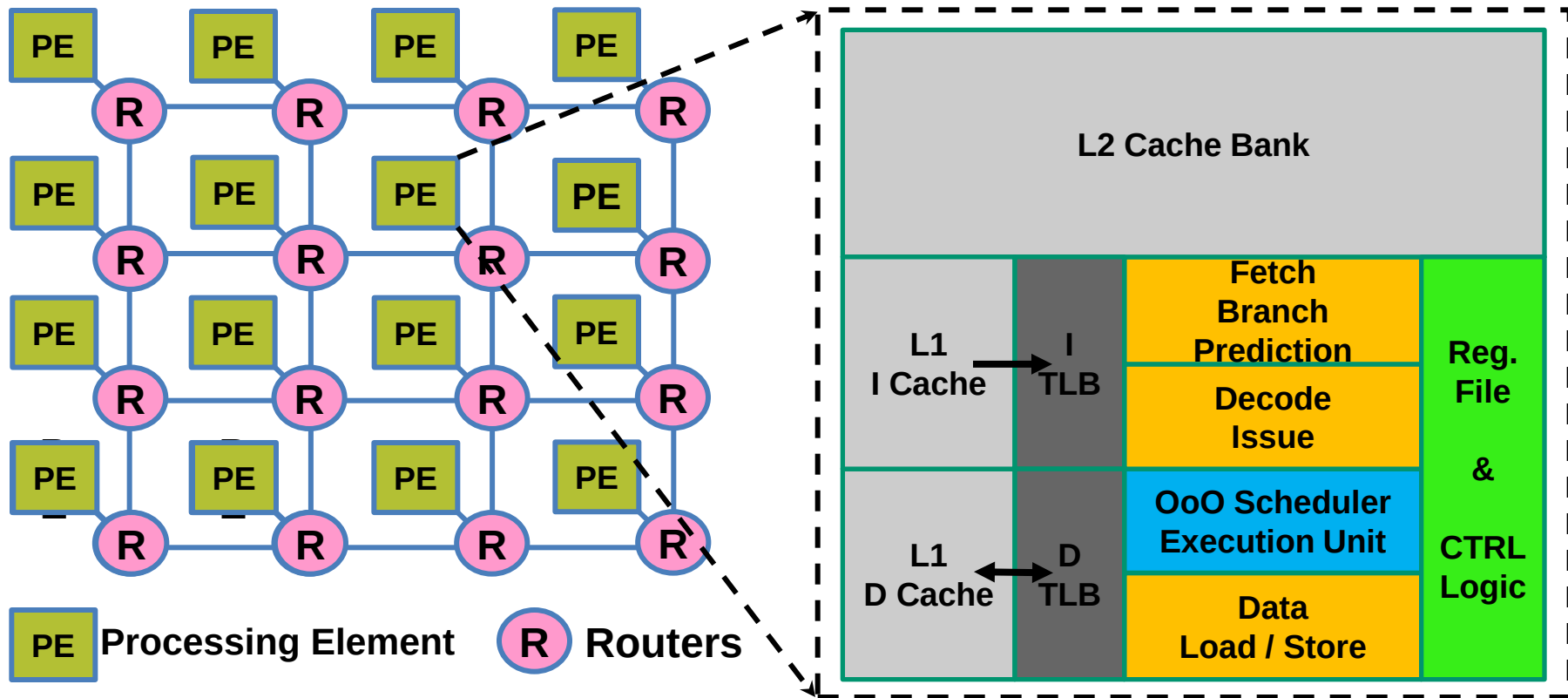
John Jose

Associate Professor

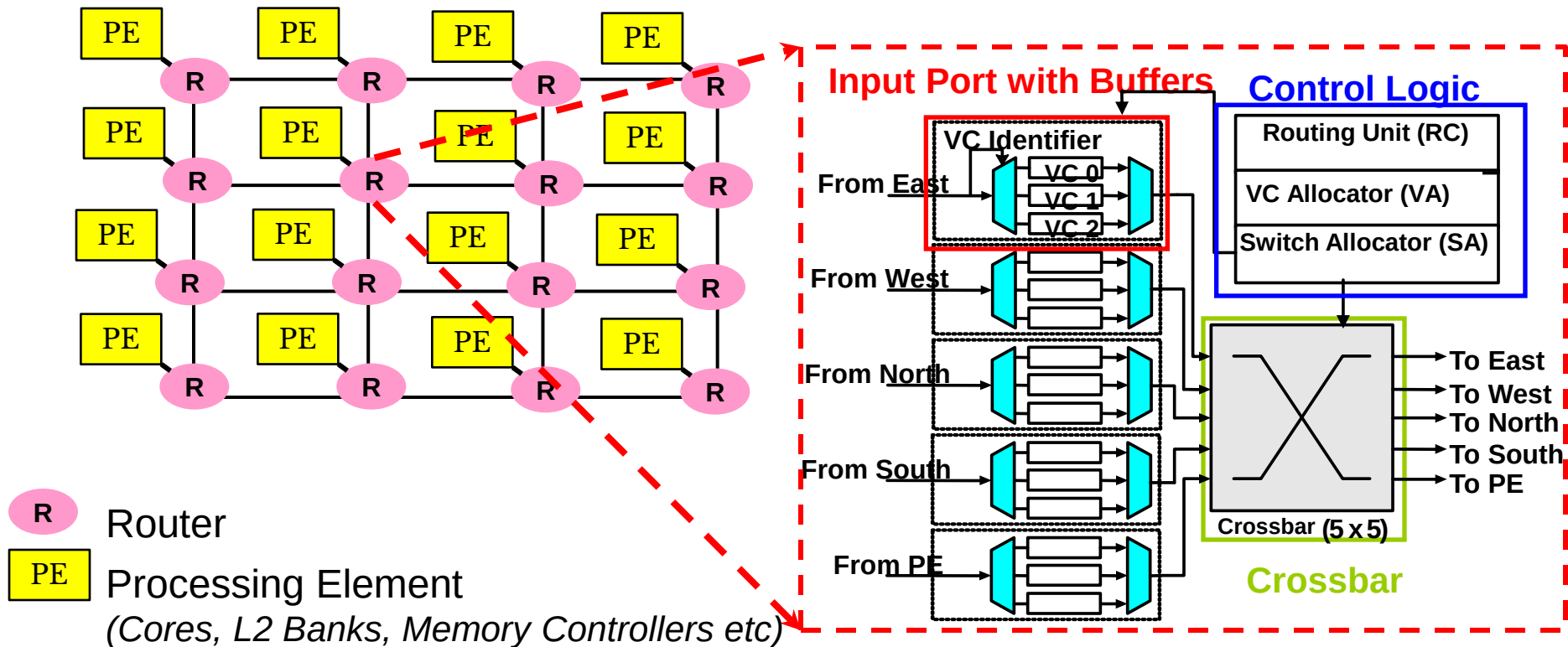
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

Tiled Chip Many-Core Processor (TCMP)

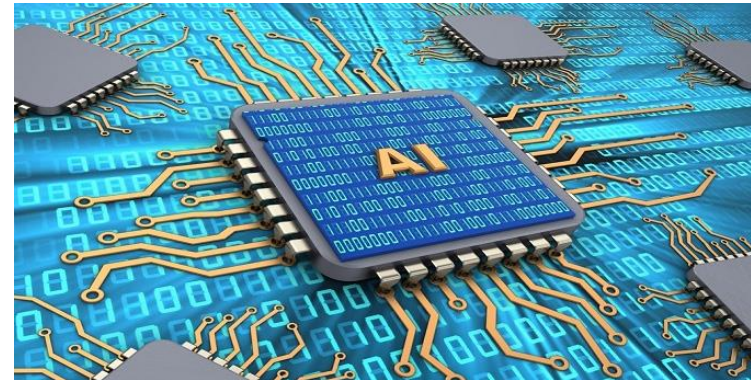


Input Buffered NoC Router



AI In Daily Life and AI on Hardware

Machine Learning And AI Is Everywhere.

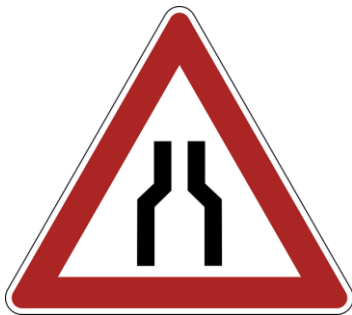


Von Neumann Bottleneck

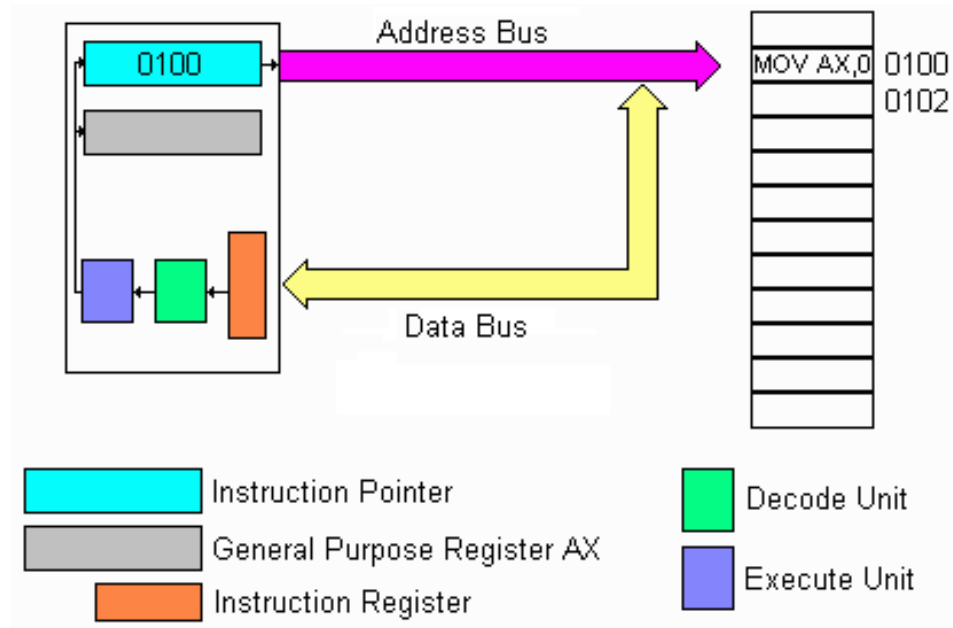
❖ Simple Fetch Cycle

- ❖ Imagine: MOV R2, [1000]
- ❖ Problem: Doing this billions of time!

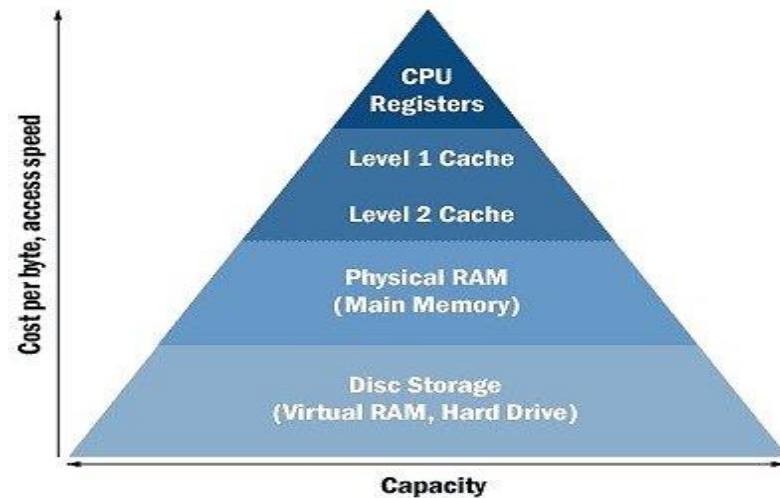
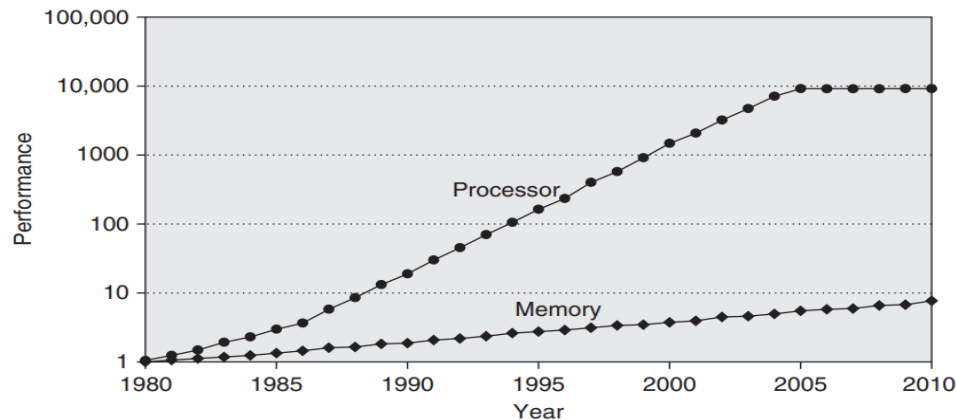
❖ Different speeds of CPU and Memory



Bottleneck



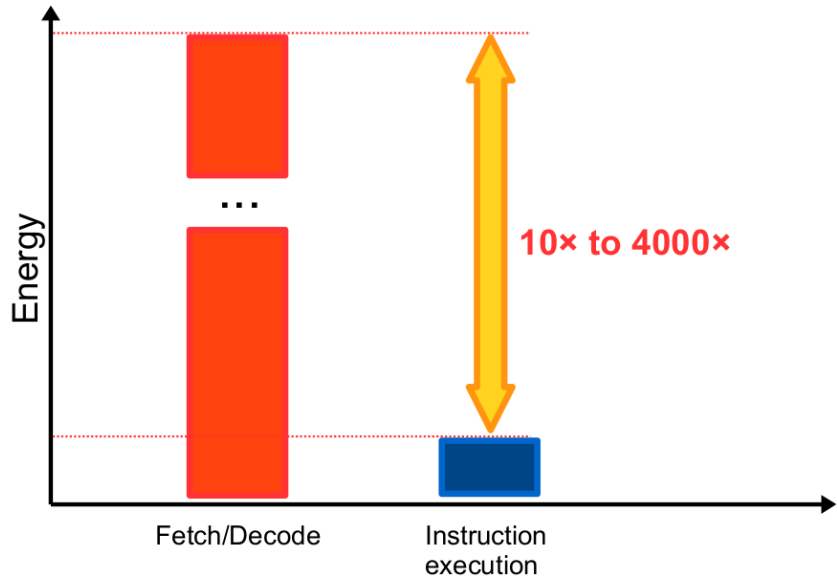
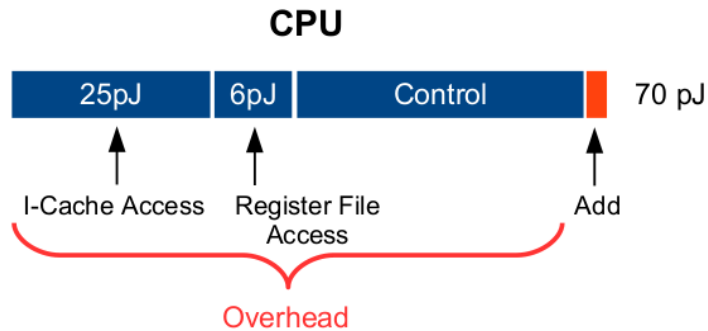
Memory Wall



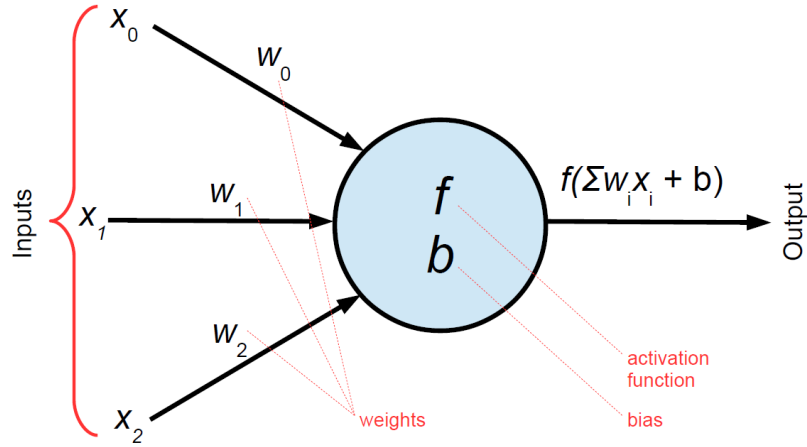
- ❖ **Different Speeds of CPU and Memory**
- ❖ Fetching anything is slow
- ❖ Doing it over and over is a nightmare !

Turing Tariff

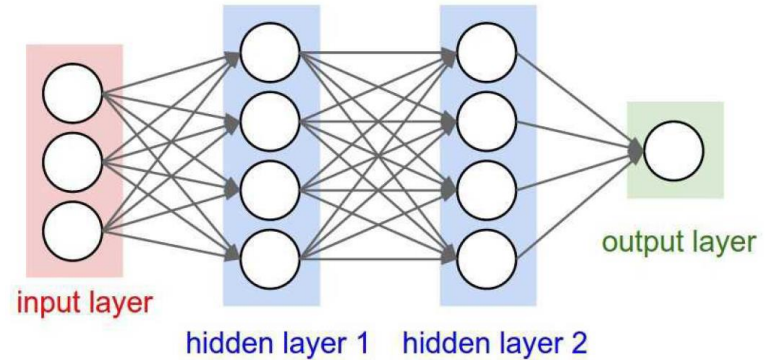
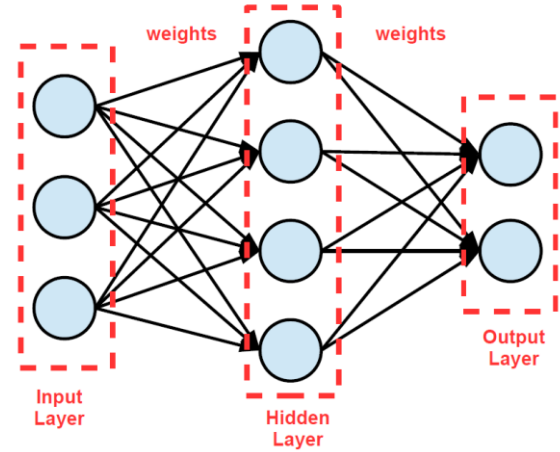
- ❖ **Cost of performing functions using General Purpose Hardware**
- ❖ GP hardware performs any functions
- ❖ Not necessarily efficiently



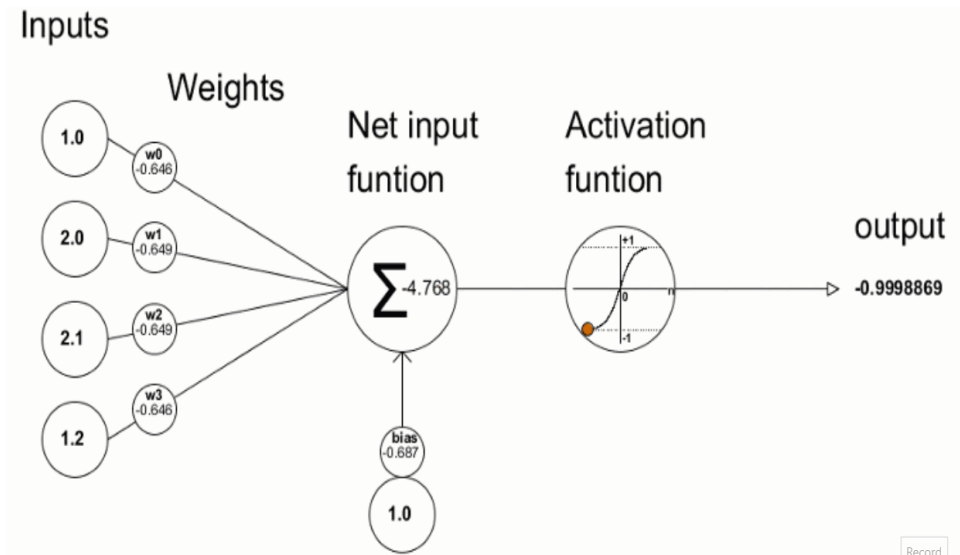
Neural Networks



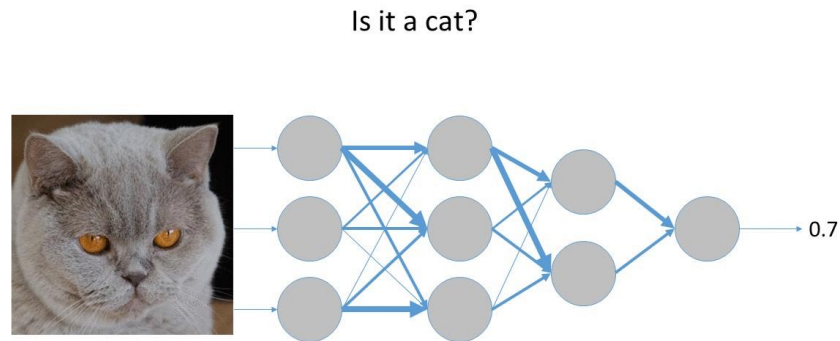
Neuron



Neural Networks

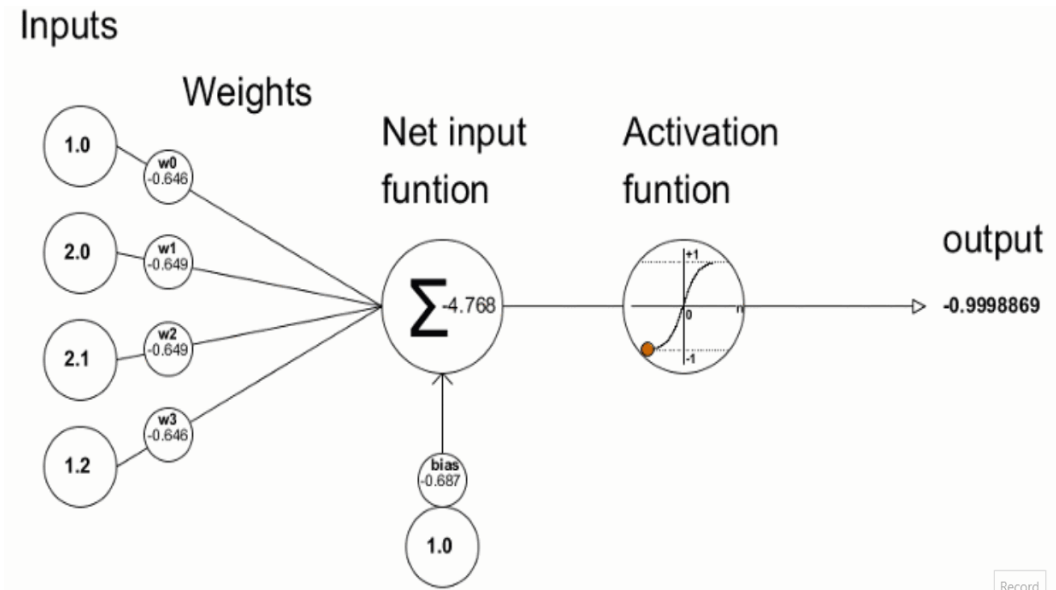


• Artificial Neuron

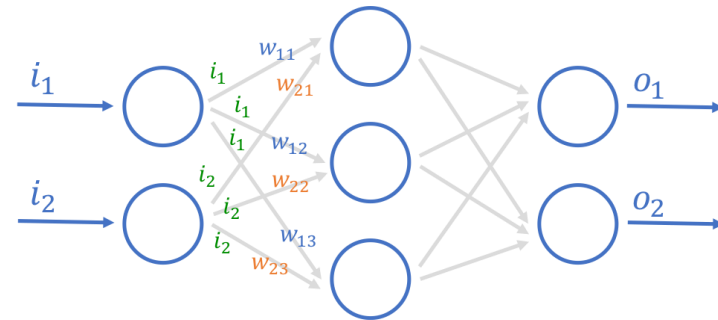


• Neural Networks

Neural Networks



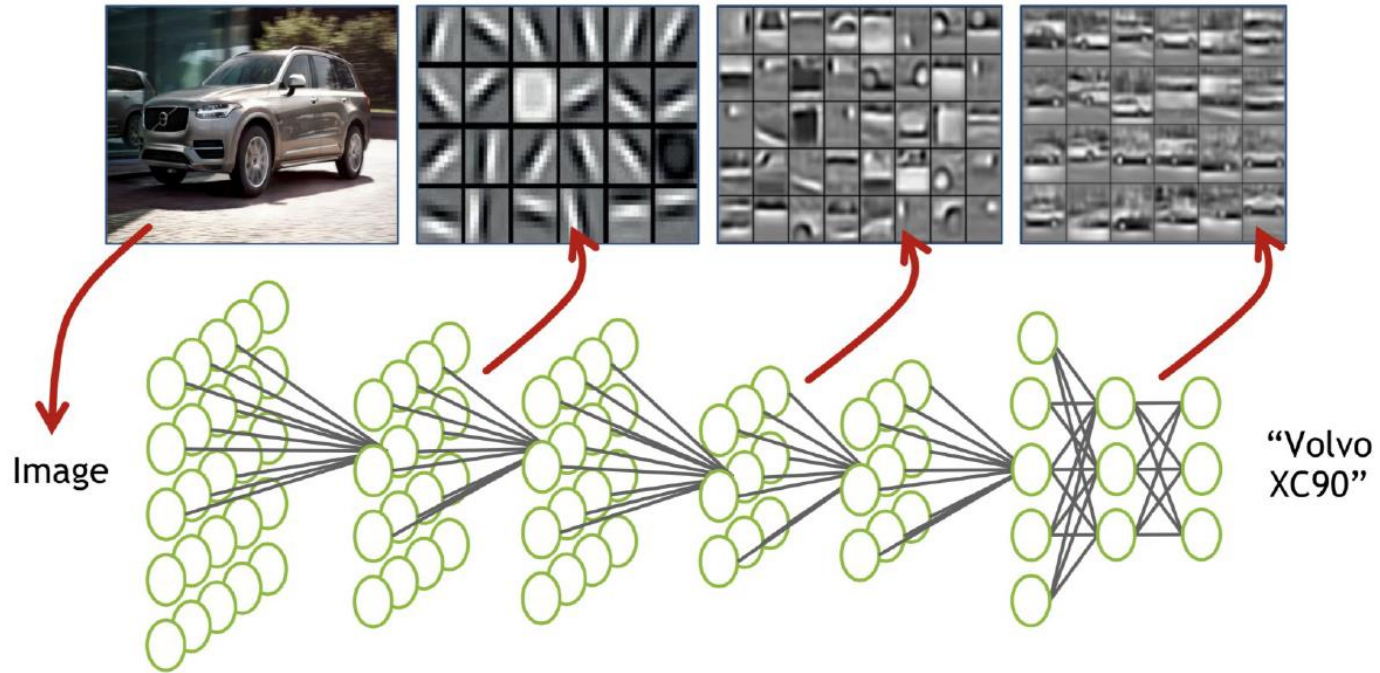
• Artificial Neuron



$$\begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} (w_{11} \times i_1) + (w_{21} \times i_2) \\ (w_{12} \times i_1) + (w_{22} \times i_2) \\ (w_{13} \times i_1) + (w_{23} \times i_2) \end{bmatrix}$$

• MAC, Matrix –Vector product

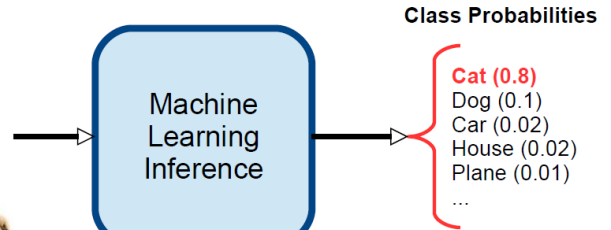
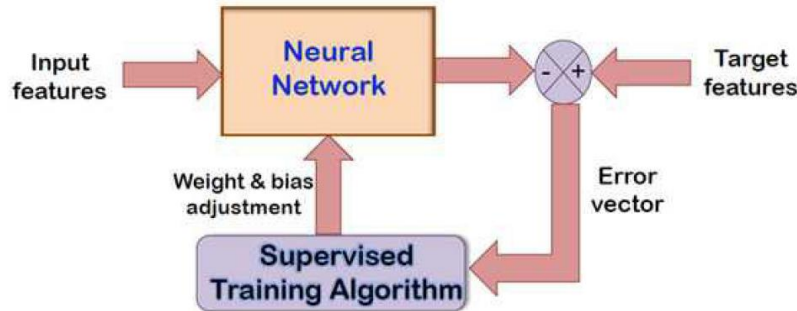
Deep Neural Network



[H. Lee et al., Unsupervised learning of hierarchical representations with convolutional deep belief networks. Communications. ACM 2011]

Deep Neural Network

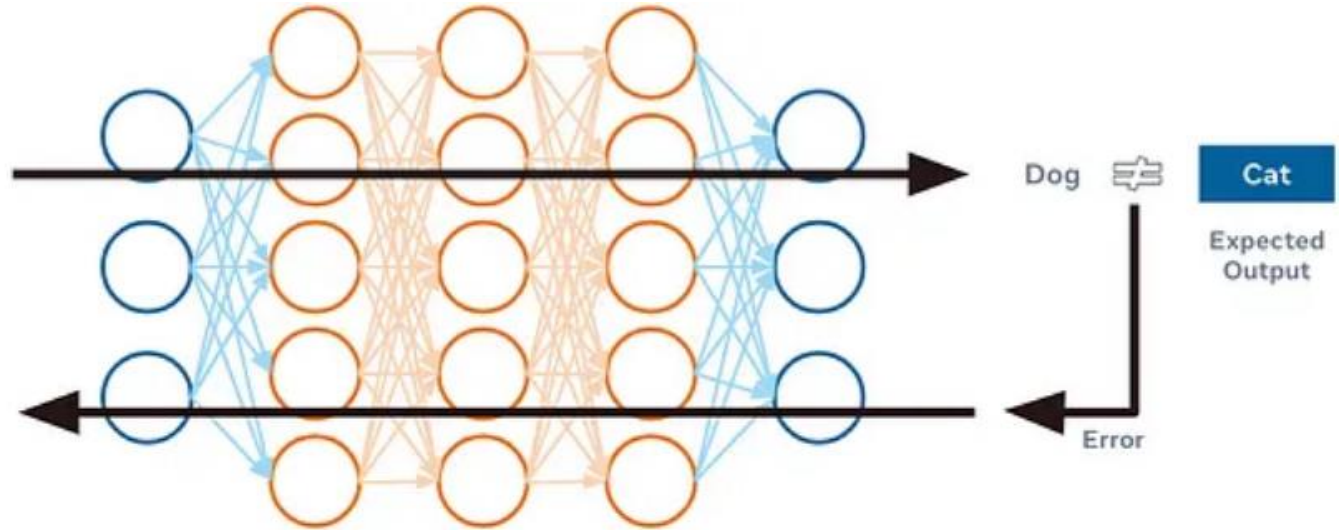
- ❖ 2 Phases – Training , Inference
- ❖ Training - Determine weights and biases
- ❖ Inference - Apply weights to determine output



Neural Network Training



Training
Input Data

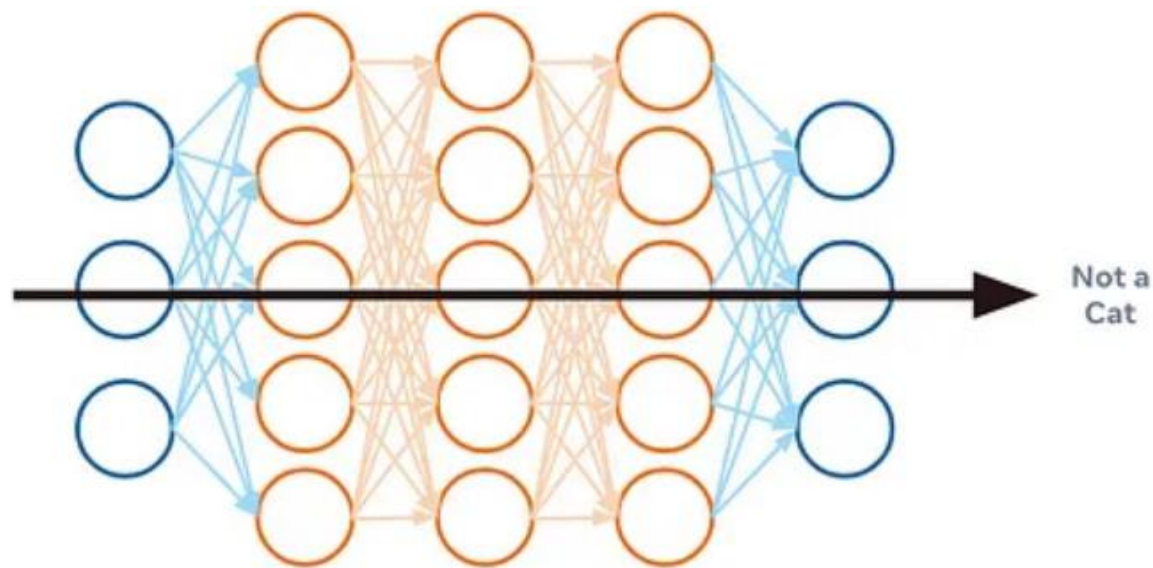


• Training

Neural Network Inference

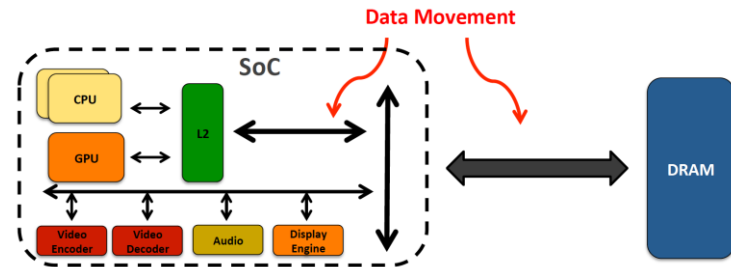
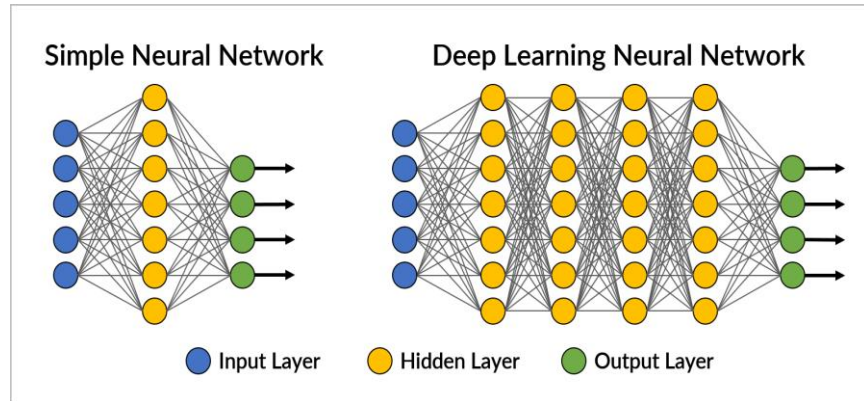
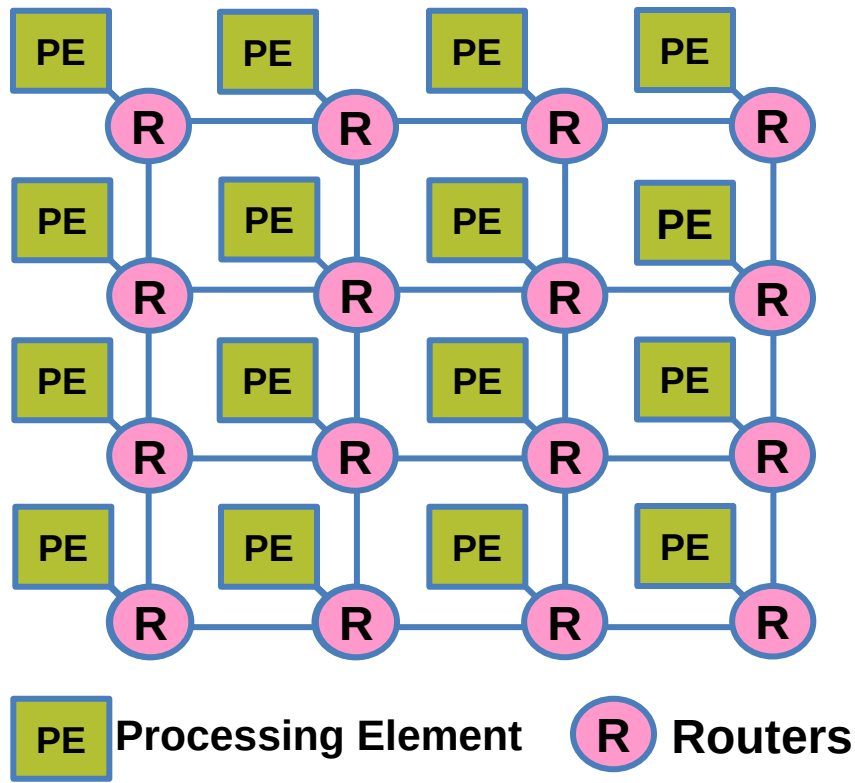


Inference
New Data



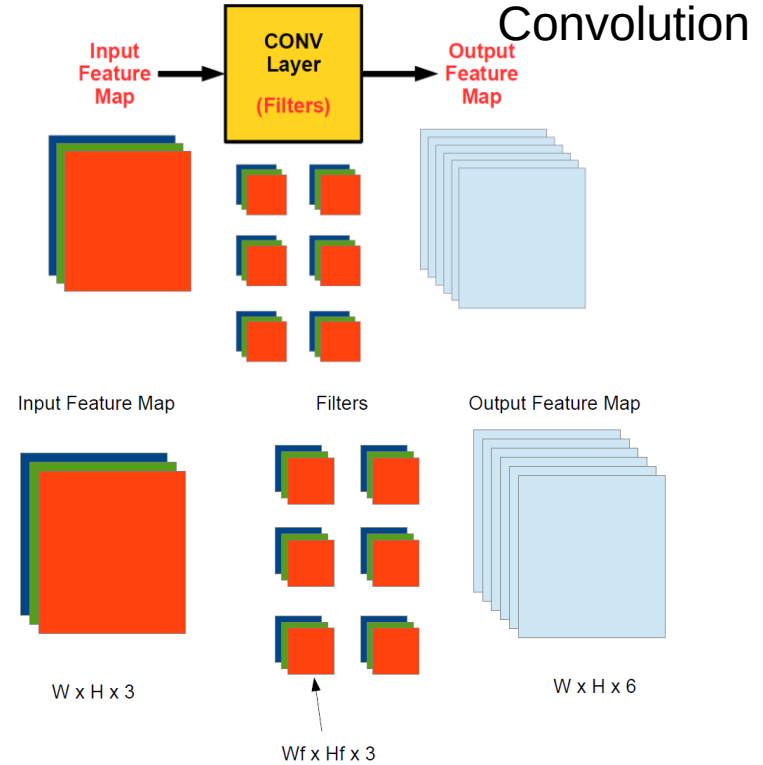
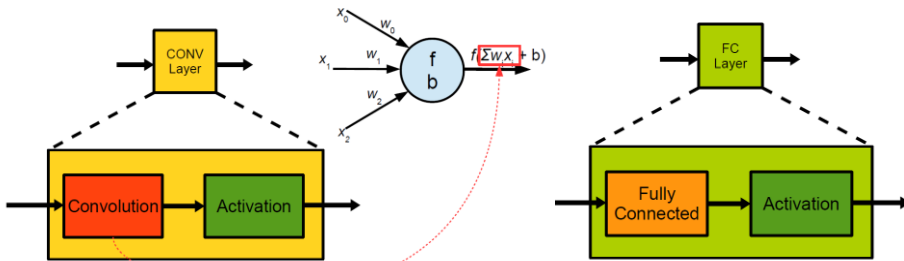
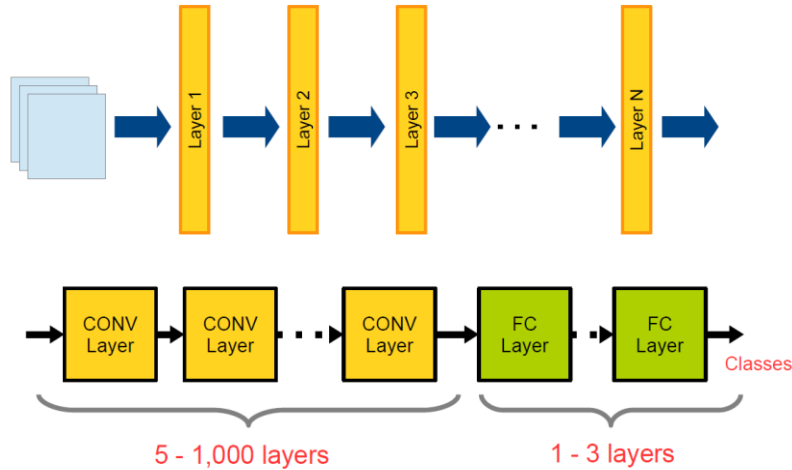
• Inference

Tiled Chip Many-Core Processor (TCMP)



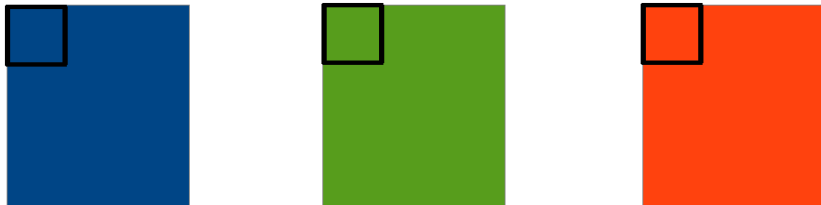
Most of the system is dedicated to storing and moving data

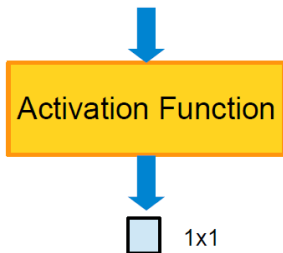
Deep Neural Network



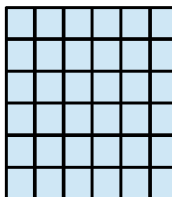
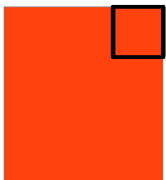
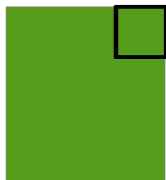
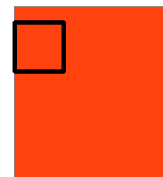
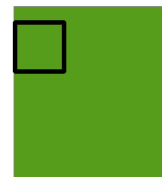
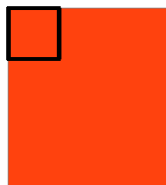
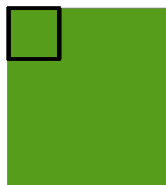
Convolutions account for more than 90% of overall computation, Dominate runtime and energy consumption

Convolution

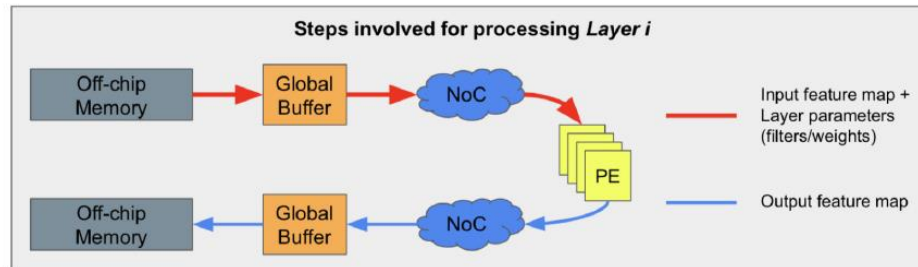
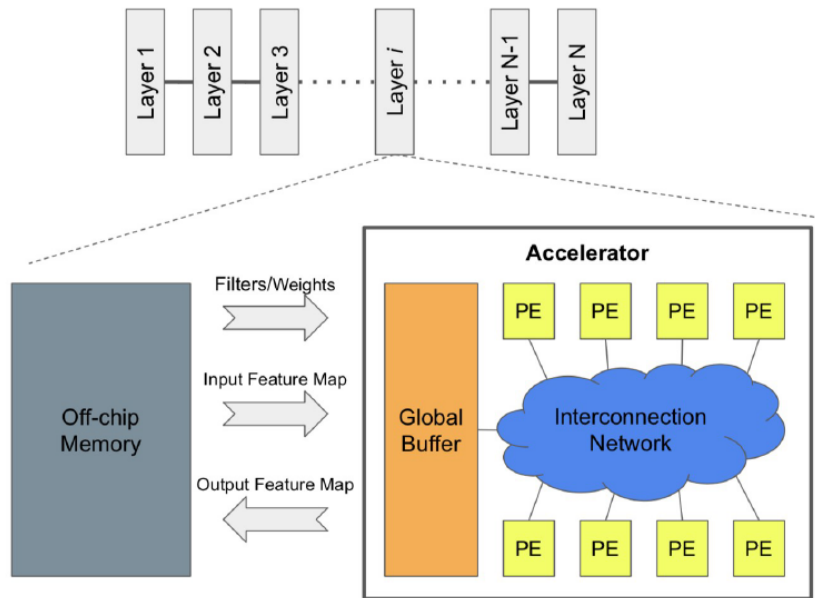


$$\begin{aligned}
 &f1[0,0,\blacksquare]*I[0,0,\blacksquare] + f1[1,0,\blacksquare]*I[1,0,\blacksquare] + \dots + f1[Wf-1,0,\blacksquare]*I[Wf-1,0,\blacksquare] + \\
 &f1[0,1,\blacksquare]*I[0,1,\blacksquare] + f1[1,1,\blacksquare]*I[1,1,\blacksquare] + \dots + f1[Wf-1,1,\blacksquare]*I[Wf-1,1,\blacksquare] + \\
 &+ \dots + \\
 &f1[0,Hf-1,\blacksquare]*I[0,Hf-1,\blacksquare] + f1[1,Hf-1,\blacksquare]*I[1,Hf-1,\blacksquare] + \dots + f1[Wf-1,Hf-1,\blacksquare]*I[Wf-1,Hf-1,\blacksquare] + \\
 &f1[0,0,\blacksquare]*I[0,0,\blacksquare] + f1[1,0,\blacksquare]*I[1,0,\blacksquare] + \dots + f1[Wf-1,0,\blacksquare]*I[Wf-1,0,\blacksquare] + \\
 &f1[0,1,\blacksquare]*I[0,1,\blacksquare] + f1[1,1,\blacksquare]*I[1,1,\blacksquare] + \dots + f1[Wf-1,1,\blacksquare]*I[Wf-1,1,\blacksquare] + \\
 &+ \dots + \\
 &f1[0,Hf-1,\blacksquare]*I[0,Hf-1,\blacksquare] + f1[1,Hf-1,\blacksquare]*I[1,Hf-1,\blacksquare] + \dots + f1[Wf-1,Hf-1,\blacksquare]*I[Wf-1,Hf-1,\blacksquare] + \\
 &f1[0,0,\blacksquare]*I[0,0,\blacksquare] + f1[1,0,\blacksquare]*I[1,0,\blacksquare] + \dots + f1[Wf-1,0,\blacksquare]*I[Wf-1,0,\blacksquare] + \\
 &f1[0,1,\blacksquare]*I[0,1,\blacksquare] + f1[1,1,\blacksquare]*I[1,1,\blacksquare] + \dots + f1[Wf-1,1,\blacksquare]*I[Wf-1,1,\blacksquare] + \\
 &+ \dots + \\
 &f1[0,Hf-1,\blacksquare]*I[0,Hf-1,\blacksquare] + f1[1,Hf-1,\blacksquare]*I[1,Hf-1,\blacksquare] + \dots + f1[Wf-1,Hf-1,\blacksquare]*I[Wf-1,Hf-1,\blacksquare] + \text{Bias}
 \end{aligned}$$


Convolution

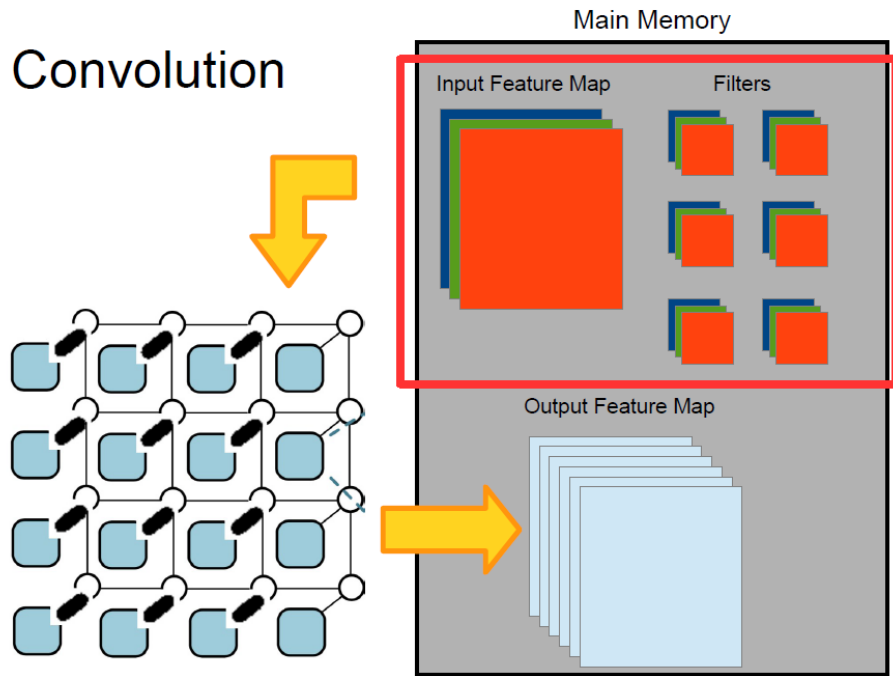


DNN Computation

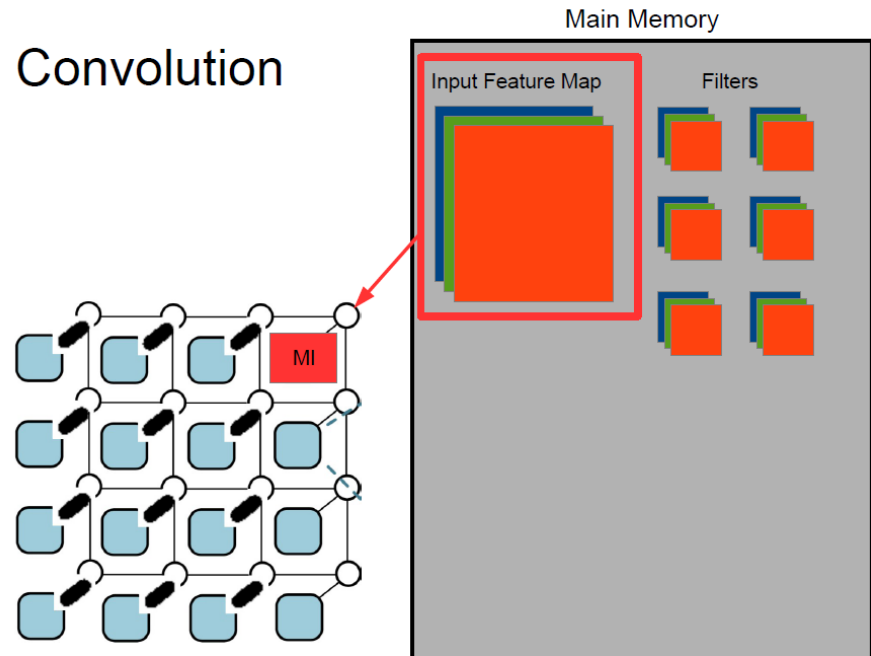


Convolution

Convolution

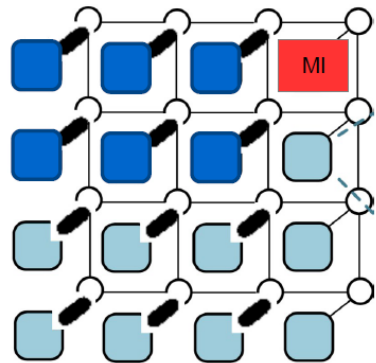


Convolution

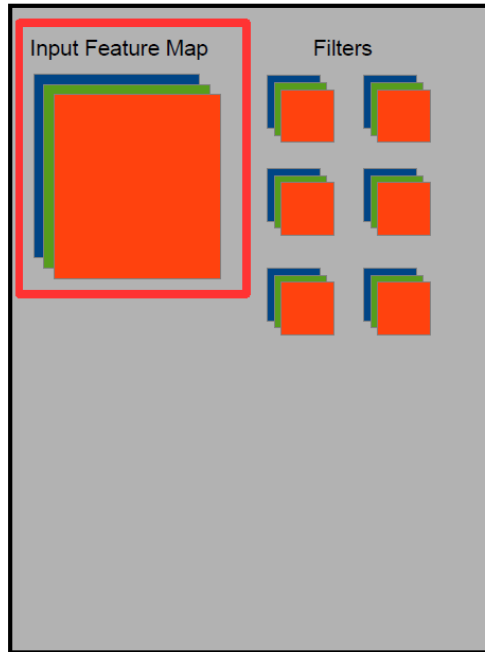


Convolution

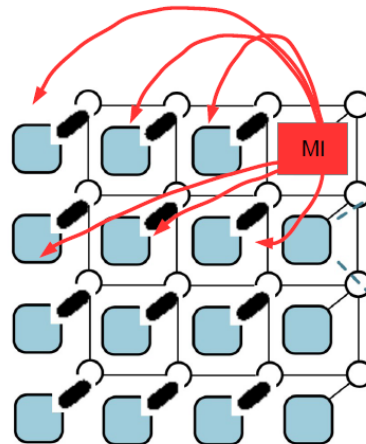
Convolution



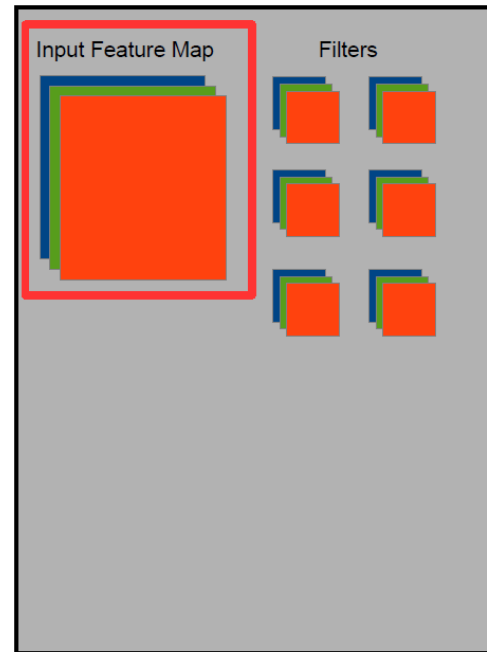
Main Memory



Convolution

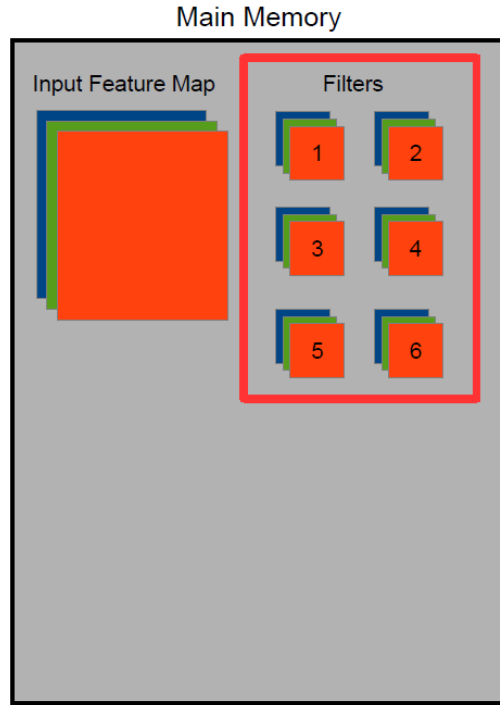
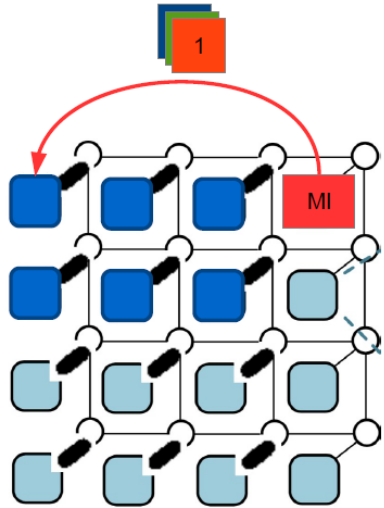


Main Memory

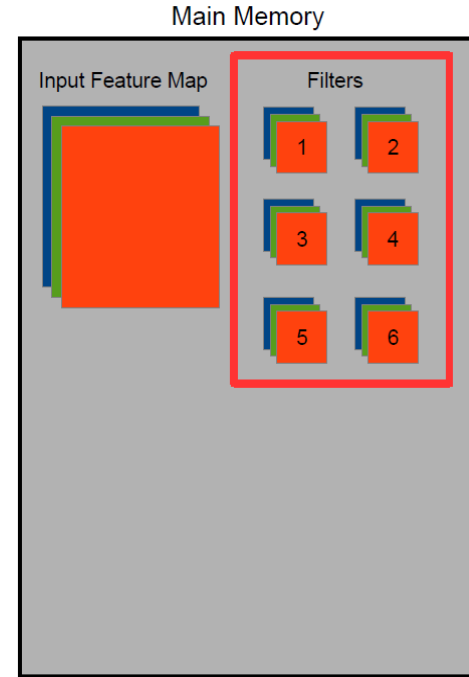
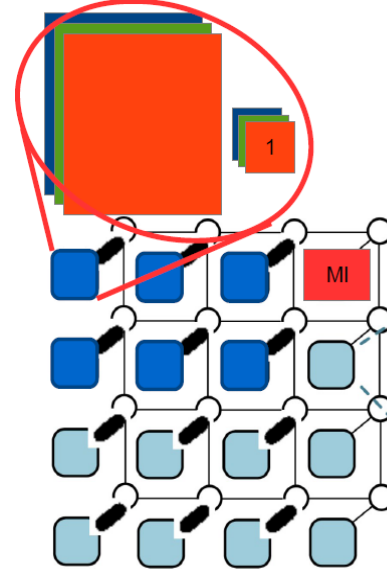


Convolution

Convolution

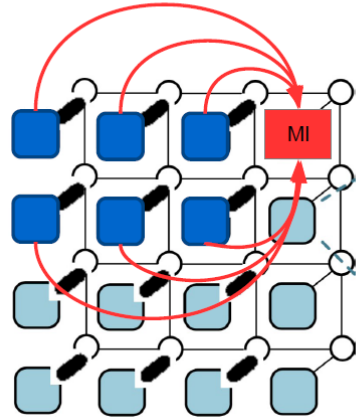


Convolution

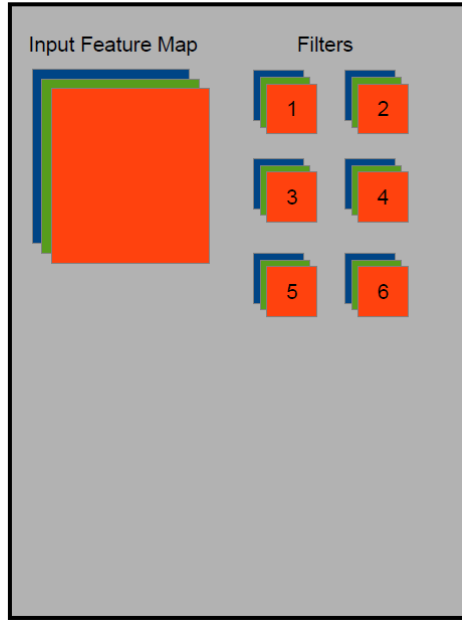


Convolution

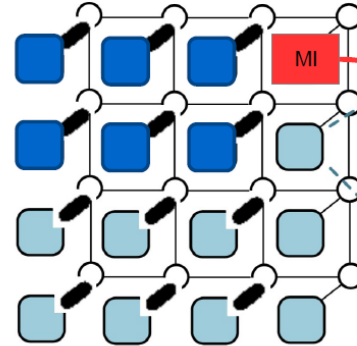
Convolution



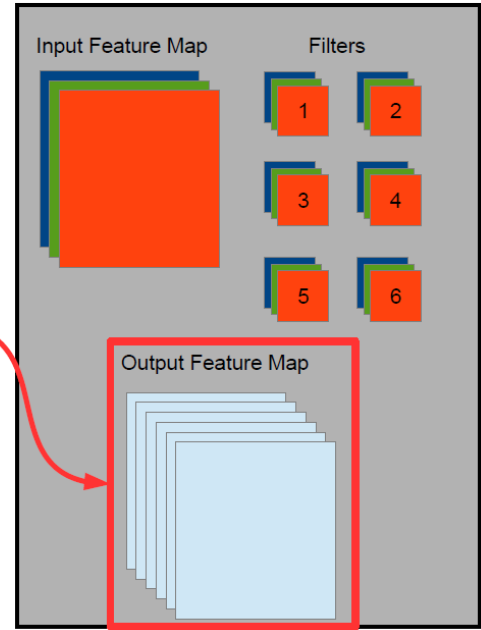
Main Memory



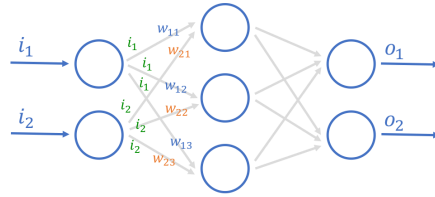
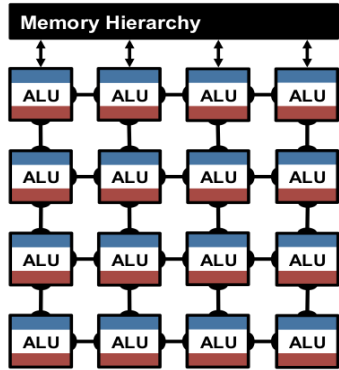
Convolution



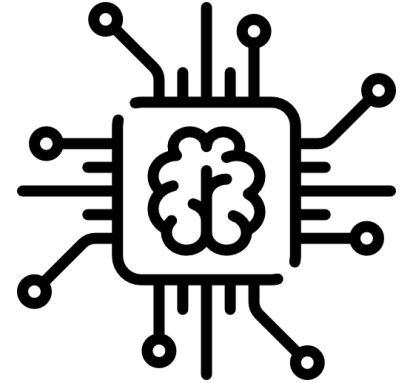
Main Memory



Neural Network Accelerators



$$\begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} (w_{11} \times i_1) + (w_{21} \times i_2) \\ (w_{12} \times i_1) + (w_{22} \times i_2) \\ (w_{13} \times i_1) + (w_{23} \times i_2) \end{bmatrix}$$



• Specialized Hardware

• Neural Networks

• NN Accelerators

Neural Network Data Flow Accelerators

❖ Dataflow Architectures

- ❖ Temporal Architectures
- ❖ Spatial Architectures

Dataflow Architectures

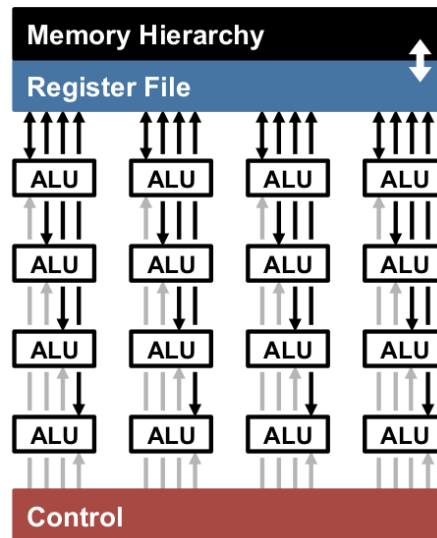
❖ Temporal Architectures

- ❖ CPUs and GPUs
- ❖ Centralized Control and Memory
- ❖ ALUs cannot talk

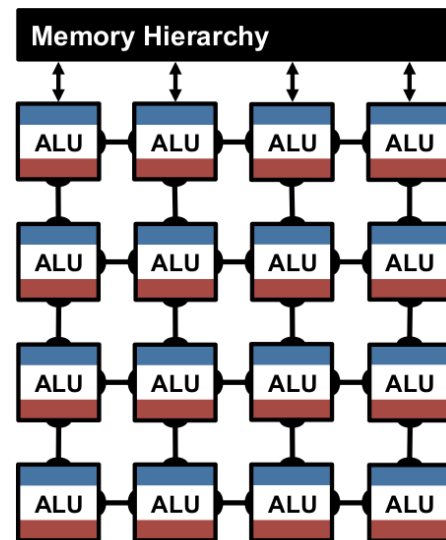
❖ Spatial Architectures

- ❖ FPGAs and ASICs
- ❖ Distributed Control and Memory
- ❖ ALUs / PEs can talk

Temporal Architecture
(SIMD/SIMT)



Spatial Architecture
(Dataflow Processing)



Dataflow Architectures

❖ Weight Stationary

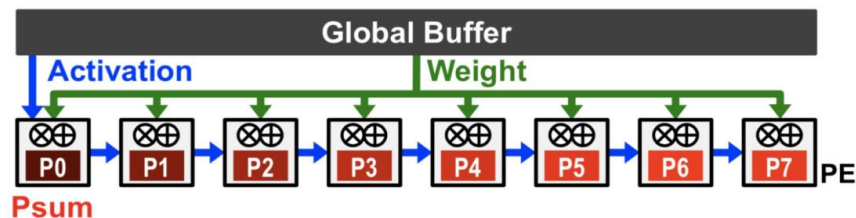
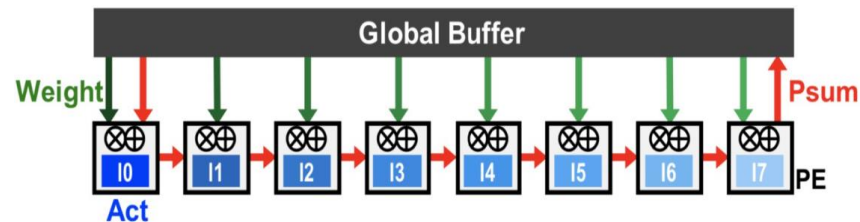
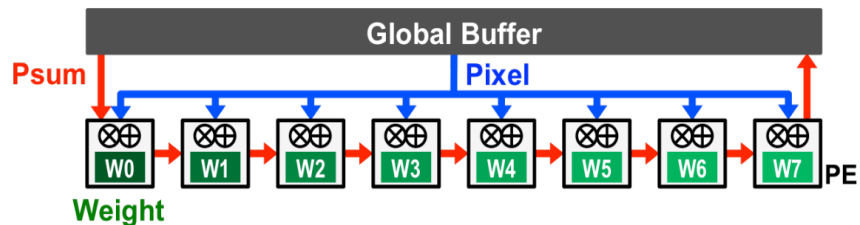
- ❖ Weight kept in PE
- ❖ Input Pixels and Partial Sums move

❖ Input Stationary

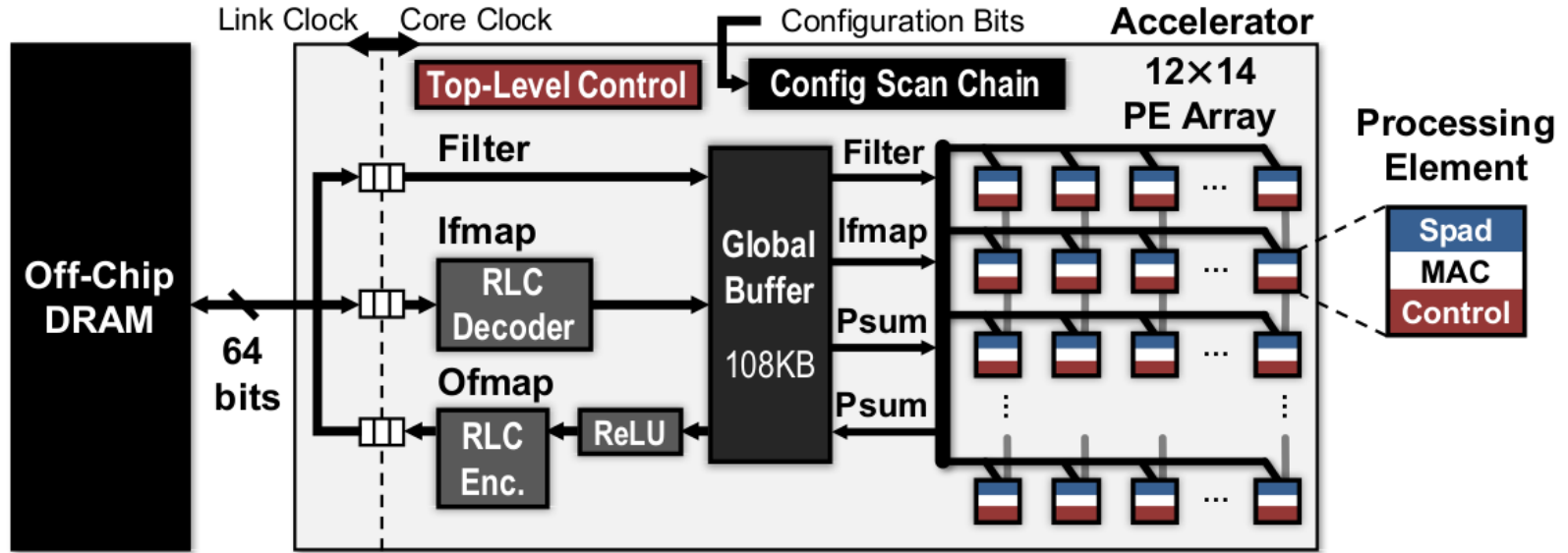
- ❖ Input kept in PE
- ❖ Weights and Partial Sums move

❖ Output Stationary

- ❖ Partial Sums kept in PE
- ❖ Input Pixels and Weights move

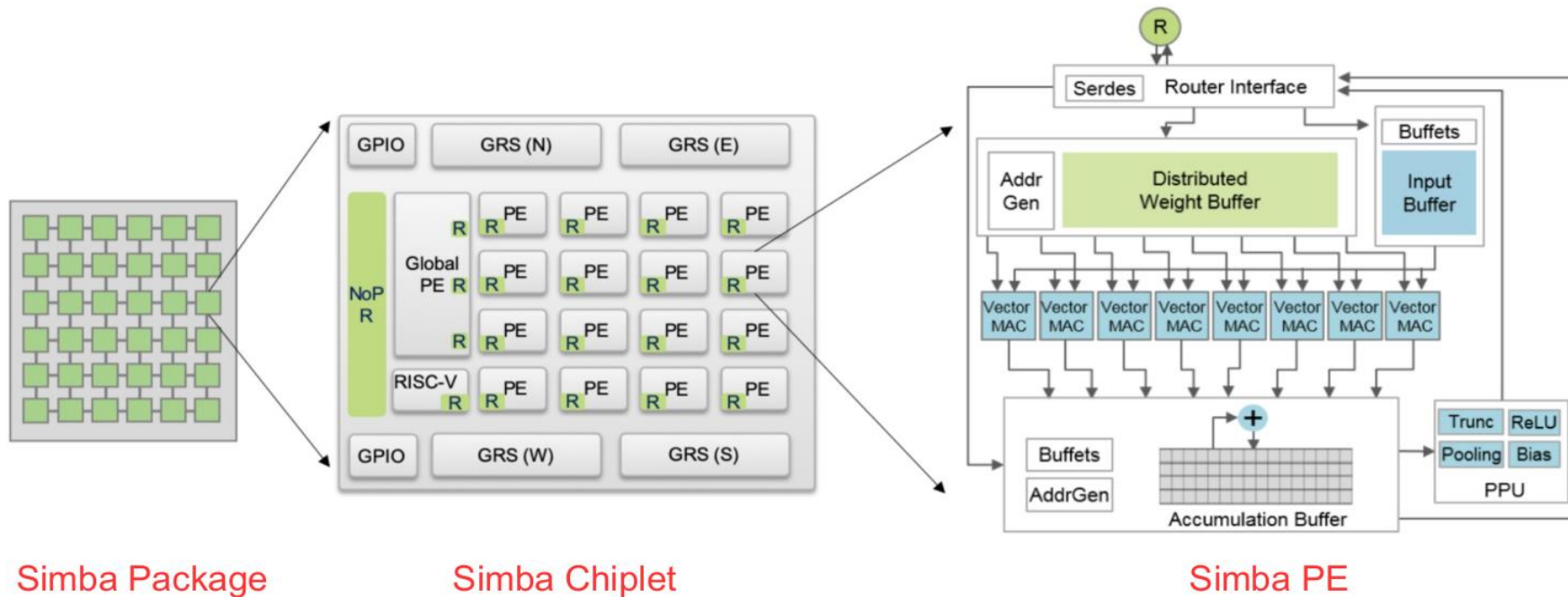


Eyeriss



• Eyeriss Accelerator

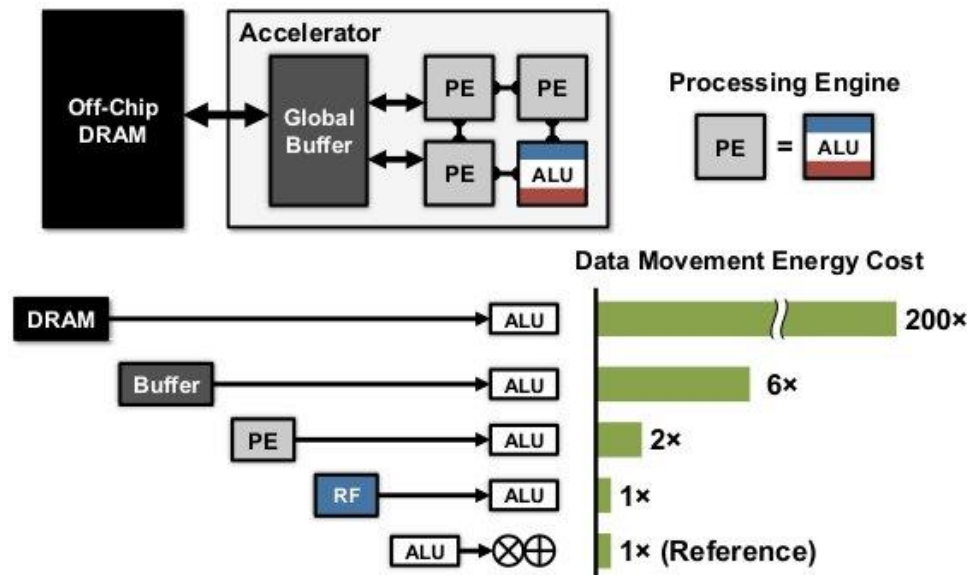
SIMBA



• Chiplet Architecture of SIMBA

Near / In- Memory Methods

- ❖ Data Movement costlier than ALU operation
- ❖ Bring data close to compute
- ❖ Physically closer – Near-Memory
 - ❖ Hybrid Memory Cube
- ❖ Same place – In-Memory
 - ❖ Resistive Crossbars



• Data Movement Cost Comparison

In-Memory Computing

- ❖ **Storage and Processing in same place**

- ❖ Data movement eliminated

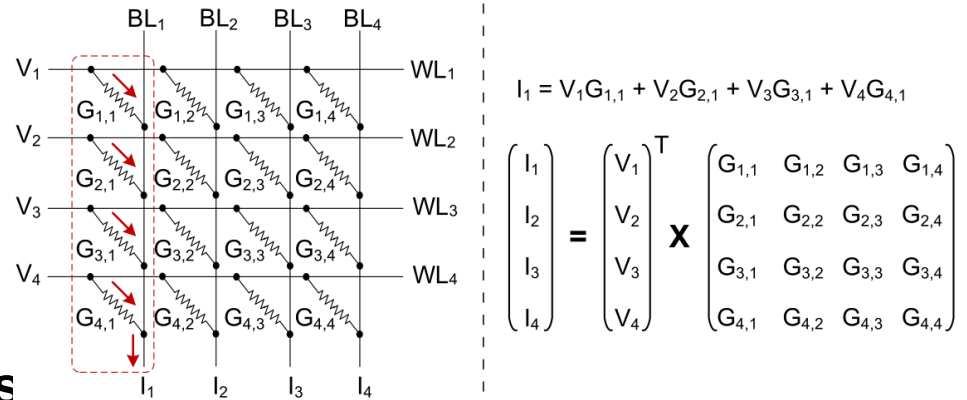
- ❖ Saves Time and Energy

- ❖ Dedicated circuits

- ❖ **In Context of Neural Networks**

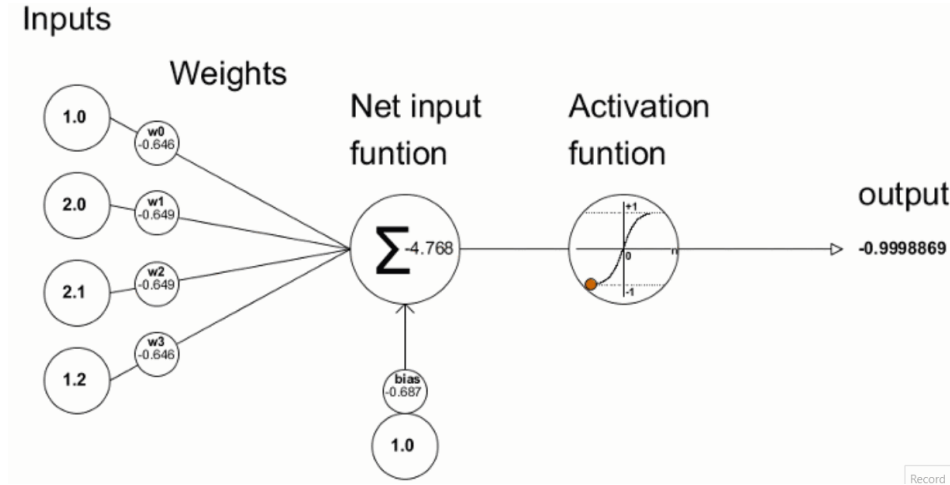
- ❖ MAC is performed

- ❖ Data / Weight is stored

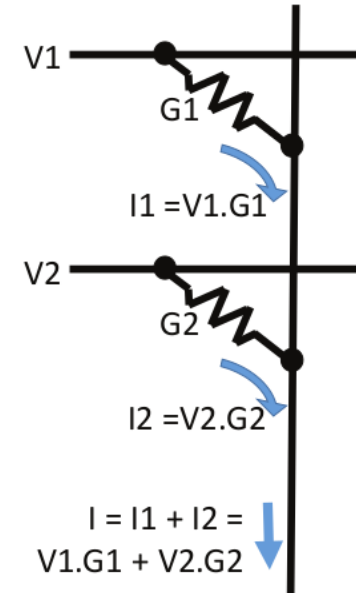


- **Memristor Crossbar for Matrix Vector Multiplication**

In-Memory Computing

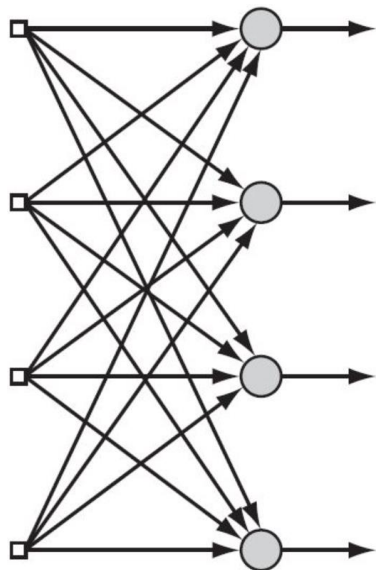


• Artificial Neuron

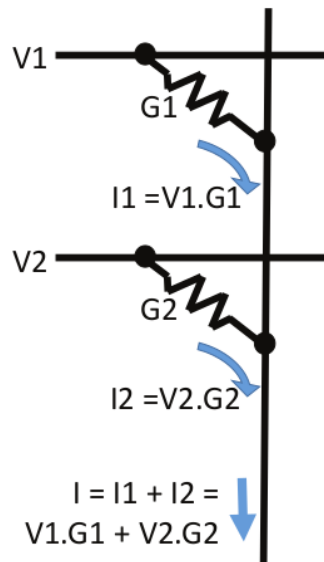


• Equivalent Circuit

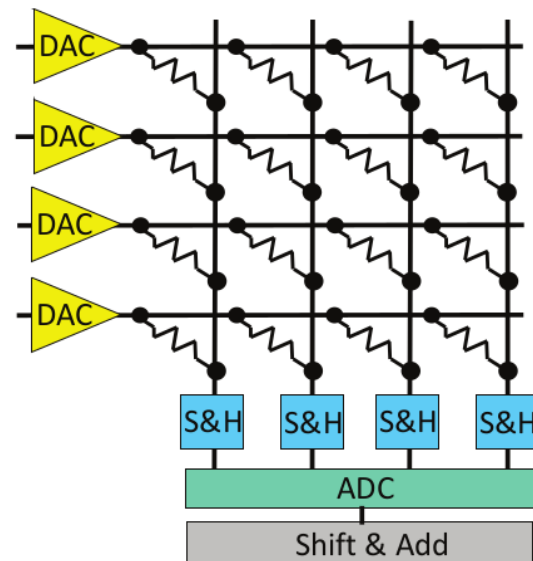
In-Memory Computing



Effectively
Weight
Stationary
Dataflow !

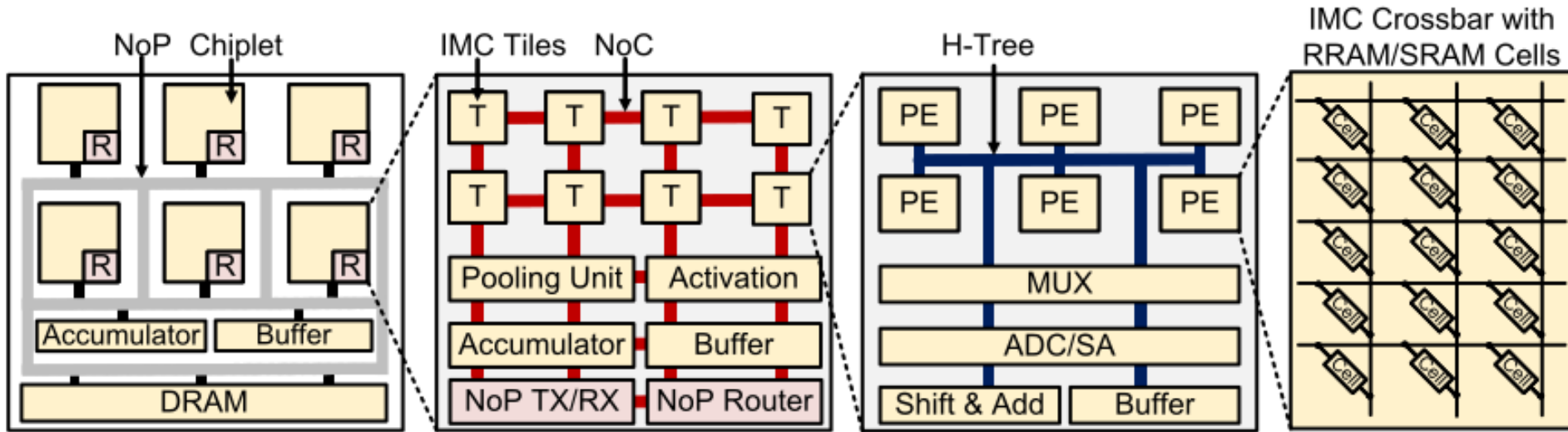


• Layer of Neural Network



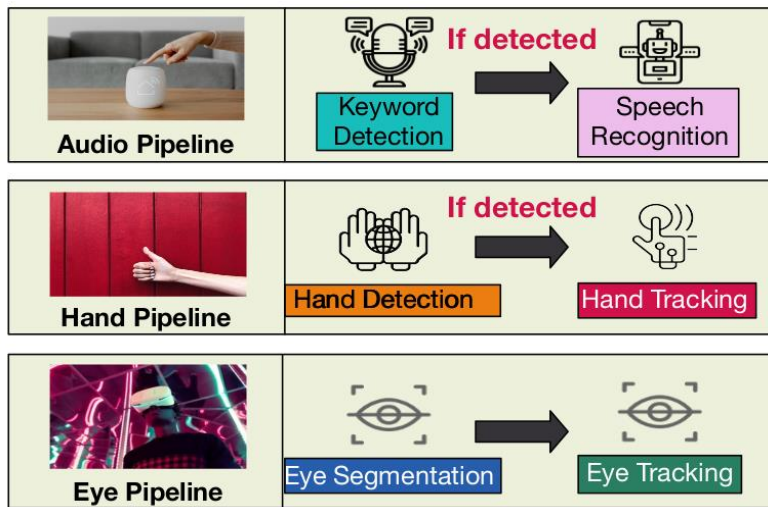
• Equivalent Circuit

SIAM



- **Chiplet architecture of SIAM**

XR Bench



Category	Task	Model
Interaction	Hand Tracking (HT)	Hand Shape/Pose (Ge et al., 2019)
	Eye Segmentation (ES)	RITNet (Chaudhary et al., 2019)
	Gaze Estimation (GE)	Eyecod (You et al., 2022)
	Keyword Detection (KD)	Key-Res-15 (Tang & Lin, 2018)
	Speech Recognition (SR)	Emformer (Shi et al., 2021)
Context Understanding	Semantic Segmentation (SS)	HRViT (Gu et al., 2022)
	Object Detection (OD)	D2Go (Meta, 2022b)
	Action Segmentation (AS)	TCN (Lea et al., 2017)
	Keyword Detection (KD)	Key-Res-15 (Tang & Lin, 2018)
	Speech Recognition (SR)	Emformer (Shi et al., 2021)
World Locking	Depth Estimation (DE)	MiDaS (Ranftl et al., 2020)
	Depth Refinement (DR)	Sparse-to-Dense (Ma & Karaman, 2018)
	Plane Detection (PD)	PlaneRCNN (Liu et al., 2019)

❖ Examples

- ❖ Multi-Task Multi-Model(MTMM)
- ❖ Real-Time execution
- ❖ QoS poses Challenge

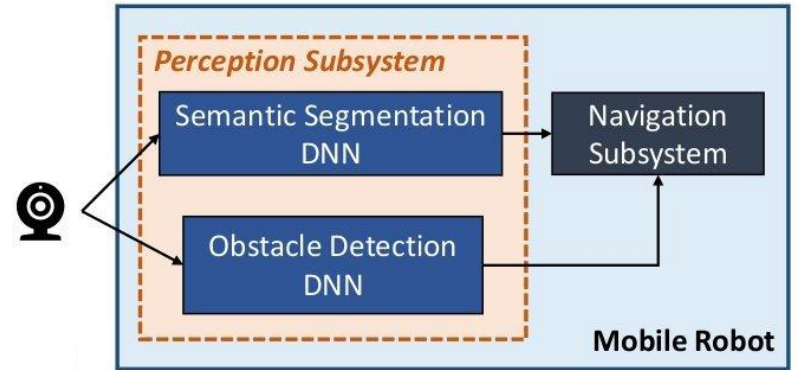
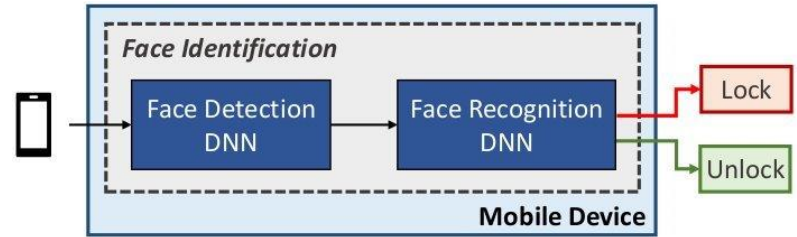
Classification of MTMM

❖ Cascade / Cas-MTMM

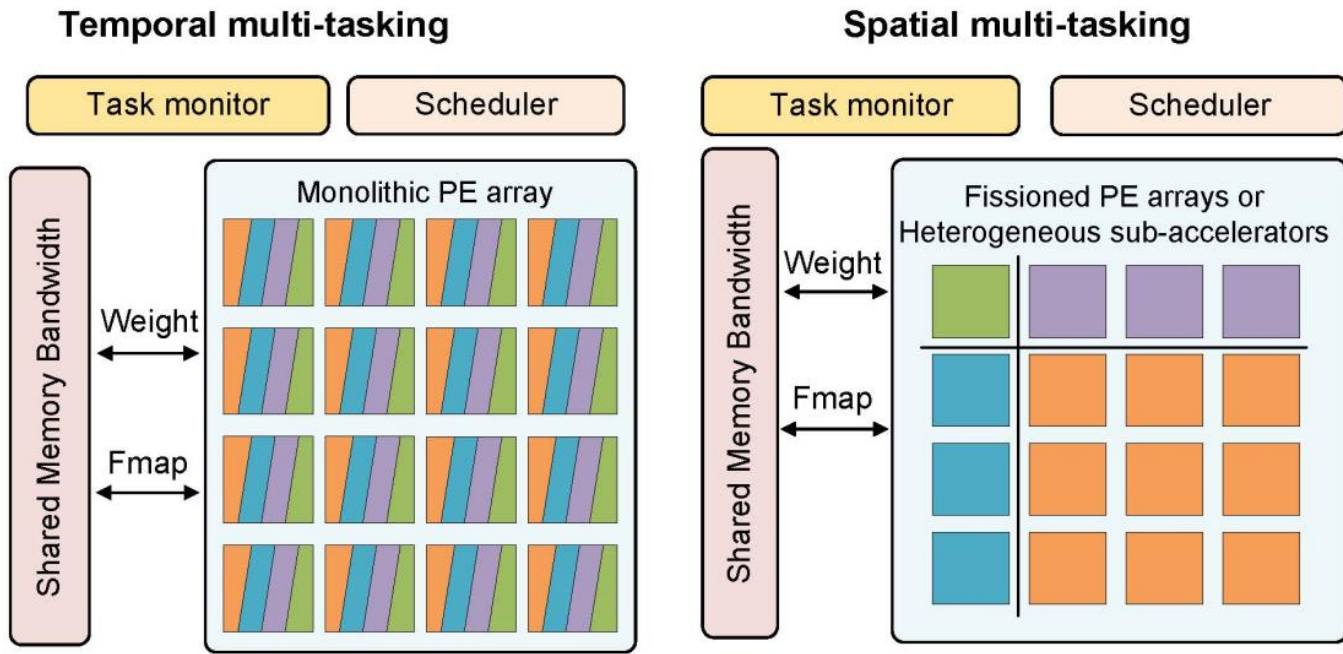
- ❖ One model after another
- ❖ Output of a model input to next

❖ Concurrent / Con-MTMM

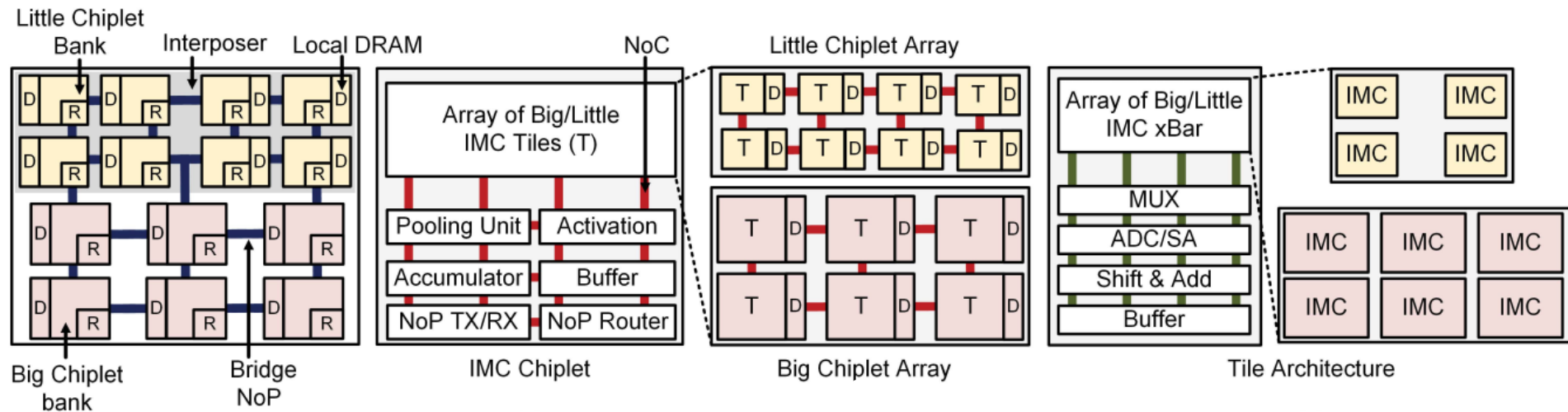
- ❖ Models execute parallelly
- ❖ Input may be same or different



HASP



Big-Little-Chiplets



Handle Data Well

- ❖ Ensure data does not overwhelm the components
 - ❖ via intelligent algorithms
 - ❖ via intelligent architectures
 - ❖ via whole system designs: algorithm-architecture-devices

Data-Centric Architectures

- ❖ **Process data where it resides**

- ❖ Processing in and near memory structures

- ❖ **Low-latency & low-energy data access**

- ❖ Low latency memory
 - ❖ Low energy memory

- ❖ **Low-cost data storage & processing**

- ❖ High capacity memory at low cost: hybrid memory, compression

- ❖ **Intelligent data management**

- ❖ Intelligent controllers handling robustness, security, cost, scaling

The Way Forward

- ❖ Data-centric system design & intelligence spread around
 - ❖ Do not center everything around traditional computation units
- ❖ Better cooperation across layers of the system
 - ❖ Careful co-design of components and layers: system/arch/device
 - ❖ Better, richer, more expressive and flexible interfaces
- ❖ Better-than-worst-case design
 - ❖ Do not optimize for the worst case, look common case
- ❖ Heterogeneity in design (specialization, asymmetry)
 - ❖ Enables a more efficient design (No one size fits all)

How to explore computer architecture ?

- ❖ Refer to IEEE/ACM transactions & journals
 - ❖ IEEE TCAD, IEEE-TVLSI, IEEE-TOC,
 - ❖ ACM-TODAES, ACM-TECS, ACM-TACO
 - ❖ JPDC, JSC, JSA, CAL, ESL
- ❖ Refer to top tier conferences
 - ❖ ISCA, HPCA, MICRO, ASPLOS, PACT, DATE, DAC, ICCAD
 - ❖ ICCD, ISVLSI, ASPDAC, VLSI-SoC, GLSVLSI, NOCS, NoCArc
 - ❖ HiPC, VLSID, VDAT, ISED

How to explore computer architecture ?

- ❖ Familiarize open source architectural simulators
 - ❖ gem5, Multi2sim, Sniper, Tejas,
 - ❖ Booksim, DRAMSim, Usimm, GPGPUSim
 - ❖ Cacti, Orion
- ❖ Model the architecture in simulators and implement them using HDLs, Verify sub-modules in FPGA kit explore further ...

Summary

- ❖ Multicore processors and on-chip clouds are going to become an integral part of future digital technologies.
- ❖ Understanding the hardware of such system will help us to design with conceptual clarity.
- ❖ Our country need good computer architects and processor design engineers with hands on exposure to VLSI design flow to cater the growing demand of skilled personnel in this domain.



johnjose@iitg.ac.in
<http://www.iitg.ac.in/johnjose/>